CHARTING THE TIMELINE OF REIONIZATION WITH A Lyα SPECTROSCOPIC STUDY

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Reionization and Galaxies in the Early Universe: Galaxies in the early universe are inherently coupled with the process of reionization, when the intergalactic medium (IGM) became transparent. The prevailing theory is that galaxies in the early universe were the primary sources of supplying ionizing photons (e.g., Finkelstein et al. 2015; Robertson et al. 2015), with faint galaxies below the detection limit of current observations contributing a significant portion of the ionizing photons (Finkelstein 2016; Livermore et al. 2017). However, constraining the cosmic star formation rate density during the epoch of reionization, which can be converted to an ionizing photon budget, is still highly controversial due to the current observational limit as well as a less constrained ionizing photon escape fraction in the galaxies (e.g., Bouwens et al. 2015; Finkelstein et al. 2015).

Understanding the duration of reionization by tracing the neutral hydrogen (HI) in the IGM provides a key constraint on how much ionizing photons from the distant galaxies were required to explain the timeline of reionization. Ionizing photon production from galaxies is tied to star formation in galaxies, thus charting the detailed time line of reionization provides a strong constraint on star formation and galaxy evolution in the early universe.

I will pursue my proposed research on tracing the time evolution of the HI fraction in the IGM during the epoch of reionization by using Lyα emission as an immediate accessible probe of reioinzation with the most comprehensive Lyα dataset at $4 \lesssim z \lesssim 8$.

Measure of the Lyα EW Distribution with the Most Comprehensive Lyα Dataset at $4 \lesssim z \lesssim 8$: In addition to all of the obtained observational data from our DEIMOS and MOSFIRE observations as described in my research summary (Figure1), I am in the VANDELS collaboration team (Co-PI's L. Pentericci and R. McLure), which provides a deep VIMOS survey of the CANDELS CDFS and UDS fields, so that I will utilize the deep optical spectroscopic dataset of > 200 LAEs at $z \sim 4 - 5$ on GOODS-S from the VANDELS survey (Pentericci et al. 2018a). This is necessary for providing the reference measure of the Lyα EW distribution from the fully-ionized universe. Plus, Pentericci et al. (2018b) published the CANDELSz7 survey data, which provides the most extensive Lyα survey dataset (> 50 LAEs) at $z \sim 6 - 7$. In collaboration with Dr. L. Pentericci, I will utilize the CANDELSz7 dataset, combined with my DEIMOS dataset, to measure the Lyα EW distribution at $z \sim 6 - 7$. Lastly, my MOSFIRE dataset is the most comprehensive dataset for Lyα search at $z > 7$. ***In summary, my dataset, which is already in-hand at $4 \lesssim z \lesssim 8$, represents the most complete spectroscopic Lyα survey ever for galaxies in/after the reionization epoch.***

My new scheme for constraining the Lyα EW distribution, which was introduced in Jung et al. (2018), is very promising as it utilizes all non-detections in the simulation step, and also it is very flexible and applicable to any spectroscopic dataset. Thus, I am already prepared for the further analysis on the additional datasets (our in-hand MOSFIRE data, the VANDELS survey, and the CANDELSz7). In the future during the fellowship years, all

available deep spectroscopic data will be combined and taken into account in my comprehensive measurement at the Lyα EW distribution. Furthermore, later at the *JWST*'s launch I will be able to extend this study easily and quickly toward even earlier universe.

Linking the Measure of the Lyα EW Distribution to Reionization: To concretely constrain the HI fraction in the IGM, we still need a linking bridge to connect the Lyα EW distribution to the HI fraction, where we need input from theoretical works (e.g., Dijkstra et al. 2014; Kakiichi et al. 2016; Mason et al. 2018), which predict the expected Lyα EW distribution depending on the IGM neutral fraction. However, the prediction is dependent on many LAE systematics as well, such as the continuum luminosity, the interstellar medium (ISM) kinematics, and the stellar mass. The first step which can be done with my dataset is understanding the dependence of the Lyα EW distribution on the continuum luminosity, or UV magnitude ($M_{\rm UV}$). With the most comprehensive dataset (as described earlier), I have an access to the extensive data at $z \sim 4-5$ from the VANDELS survey and will be able to constrain the Lyα EW distribution as a function of $M_{\rm UV}$, for instance between bright ($M_{\rm UV} < -20.25$) and faint ($M_{\rm UV} > -20.25$) populations in the ionized universe at $z < 6$. This will give us the $M_{\rm UV}$-controlled reference measure in the fully-ionized universe for a comparison to the measure of the Lyα EW distribution at $z \sim 6-8$ where the IGM contains neutral hydrogen.

Furthermore, I will aggressively keep proposing additional Keck/MOSFIRE telescope time (through NASA allocation) to detect other rest-UV emission lines (e.g., CIII]) as well as Lyα lines at $z > 6$. The other rest-UV detections will allow us to examine the ISM properties, which is necessary information to deal with the LAE systematics (e.g., Stark et al. 2017).

Additional Key Outcome – Testing Photometric Redshifts with Spectroscopic Redshifts: As spectroscopic confirmation of galaxies at $z \gtrsim 6$ is observationally expensive, photometric redshift selection provides a powerful means to construct extensive high-redshift galaxy catalogs based on multi-wavelength imaging survey data (e.g., Stark et al. 2009; Bouwens et al. 2015; Finkelstein et al. 2015). However, photometric redshift measurements alone cannot completely remove the possibility that high-redshift candidate galaxies can be low-redshift interlopers. Thus, statistical studies using photometric redshifts could be biased if there are uncovered systematic effects in photometric redshifts. Interestingly, a recent spectroscopic confirmation of Lyα emission at z =7.452 from the *HST* Faint Infrared Grism Survey (FIGS) (PI: S. Malhotra; Pirzkal et al. 2017) deviates from photometric redshift at $\sim 2\sigma$ level (Larson et al. 2018). Additionally, in my DEIMOS paper, some of the Lyα emission lines lie at the wings of the photometric redshift distributions, implying that perhaps the photometric redshift uncertainties are underestimated. Also, as essentially every quantity I derive (UV luminosity, stellar mass, star-formation rate) is dependent on the redshift accuracy, improving these estimates of the uncertainties is crucial. However, a comprehensive analysis on the accuracy of the photometric redshift measurements, specifically a calibration of these probability distribution functions, at $z \gtrsim 6$ is still lacking. What is needed is a large number of spectroscopic redshifts, which is what my MOSFIRE dataset contains. For example, preliminary results of Lyα search from my MOSFIRE data shows $\gtrsim 7$ Lyα detections at $z \gtrsim 7$ (Figure 3). By performing this analysis with my dataset, which will be part of my proposed research, I will contribute to future studies with both *HST* and *JWST* by optimizing future photometric redshift selection techniques.

Fellowship Timeline: I am already involved in the proposed project. My previous publication (Jung et al. 2018) introduces my new scheme for constraining the Lyα EW distribution

with the result of our DEIMOS data, and the MOSFIRE data reduction has been completed, ready to be incorporated in the next publication. To successfully accomplish the proposed research goals in my fellowship years, I will write three publications from the proposed project after finishing my dissertation. The detailed timeline is described below.

Year 0: Prior to the fellowship – **Finish the first analysis on the deepest MOSFIRE spectroscopic data for six candidate galaxies (Figure 4), reporting a new Lyα emission at** $z = 7.6$, which will be submitted soon.

Year 1: The MOSFIRE data reduction has been completed, and **I will report all new Lyα detections at at $z > 7$ in the first publication** during the fellowship years from the fully-reduced MOSFIRE observations for 72 candidate galaxies on GOODS-N, publishing the most comprehensive dataset at $z \sim 7 - 8$ (Figure 3).

Year 2: Analyze > 200 LAEs at $z \sim 4 - 5$ from the VANDELS data and **write the second paper on the continuum luminosity dependence of the Lyα EW distribution in the ionized universe at** $4 \lesssim z \lesssim 5$. This will bring a reference measurement of the Lyα EW distribution to be compared with the Lyα EW distribution at $z > 7$ when the IGM in the universe is significantly neutral.

Year 3: I will write **the third paper on the detailed timeline of reionzation** at $z \sim 6$ – 8, combined with the model predictions of reionization on the expected evolutions of the IGM and the Lyα EW distribution (e.g., Dijkstra et al. 2014, Kakiichi et al. 2016, Mason et al. 2018), providing a very strong observational constraint for deriving a necessary ionizing photon budget from the dominant ionizing sources, thus galaxies.

Relevance to STScI: My proposed research is perfectly aligned with the science goals of the Space Telescope Science Institute as it was originated from *HST* CANDELS survey for investigating the early universe, and I will extended this study with *JWST*. My proposed study provides a necessary tool for analyzing the future *JWST*/NIRSpec spectra. I am already closely collaborating with the STScI scientists in my PhD project, including the CANDELS team. Also, as a member of the accepted *JWST* ERS program (CEERS, PI: S. Finkelstein), I have worked together with the CEERS team for planing my *JWST* science, which will be expedited during my fellowship years at STScI.

Concluding Remarks: With my research experience, which covers both theoretical and observational contexts (analyzing spectroscopic & photometric data from ground-based & space telescopes as well as theoretical models from numerical simulations & semi-analytical models) on related research areas, I have trained myself to be well-prepared for my proposed research, being equipped with necessary skills. I will pursue my proposed research on tracing the time evolution of the HI fraction in the IGM during the epoch of reionization with using Lyα emission as an immediate accessible probe of reionization. With the *JWST* launch, I will be prepared for incorporating the extensive data of Lyα observations in the early universe to chart a full history of reionization in the last year of my fellowship years. The proposed research on Lyα emission as a probe of reionization is very timely, as the first HETDEX (Hobby-Eberly Telescope Dark Energy Experiment) data is coming out and expected to provide extensive studies on LAE systematics in near future; HETDEX utilizes the Visible Integral-field Replicable Unit Spectrograph (VIRUS) to provide blind, wide-area, integral field spectroscopic survey, and will provide $\sim$ 1M LAE detections at $1.9 < z < 3.5$. I have my collaborators in the HETDEX science team (e.g., Dr. S. Finkelstein), thus I will keep pursuing more comprehensive understanding of Lyα systematics, which is necessary to interpret Lyα emission as a probe of reionization.

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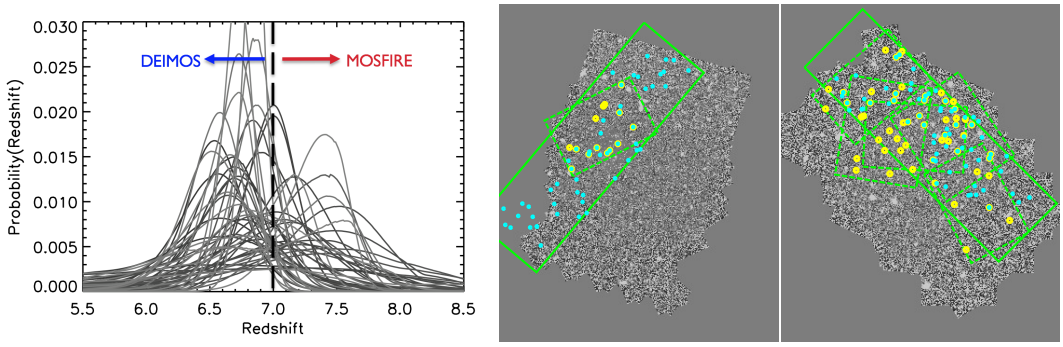

Figure 1. (Left) the redshift probability distribution functions of $z_{phot} \sim 7$ galaxies. Even though the maximum probabilities are arount at $z \approx 7$, those galaxies may be located at any redshift between $z = 6$ and $z = 8$. Either one of DEIMOS and MOSFIRE can cover the entire redshift range, thus it is critical to utilize both instruments. (Right) mask designs of spectroscopic observational survey in the GOODS-S (left) and GOODS-N (right) fields. Observed areas are marked by green rectangles; larger one is DEIMOS (5'x16.7'), and smaller dashed one is MOSFIRE (6'x4'). Cyan and yellow circles are observed galaxies with DIEMOS and MOSFIRE, respectively.

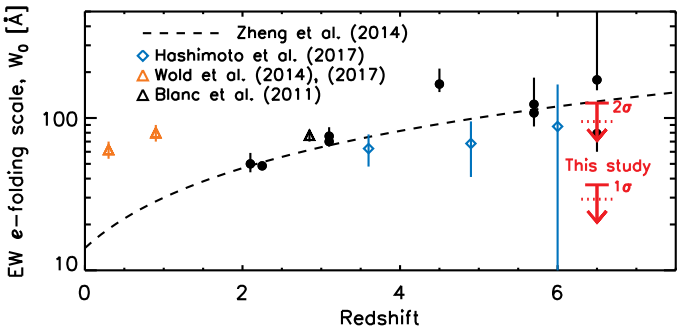

FIGURE 2. (Figure 8 of Jung et al. 2018) the redshift dependence of the Lyα EW e-folding scale (W_0). My measure of the Lyα EW distribution at $6.0 < z < 7.0$ rejects the Lyα EW distribution with $W_0 > 36.4$Å (125.3Å) at 1σ (2σ) significance (shown as red downward arrows). This provides additional evidence that the EW distribution of Lyα declines at $z > 6$, suggesting an increasing fraction of HI in the IGM at that epoch.

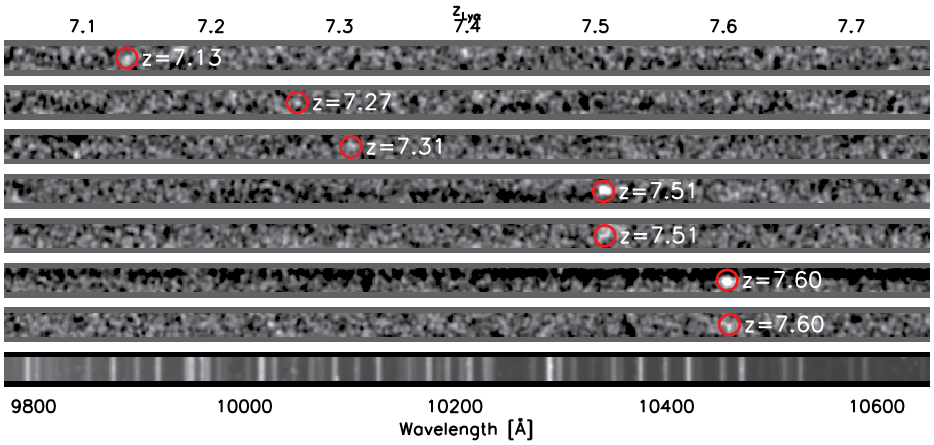

FIGURE 3. Preliminary results from 19 target galaxies in our MOSFIRE observations with $\gtrsim$10hr integration time suggests at least $\gtrsim 7$ Lyα detections at $z > 7$. The figure displays seven example Lyα emission lines detected at $z > 7$. We will increase the number of detections, analyzing the entire MOSFIRE dataset for 72 target galaxies. The bottom row shows sky-emission lines.

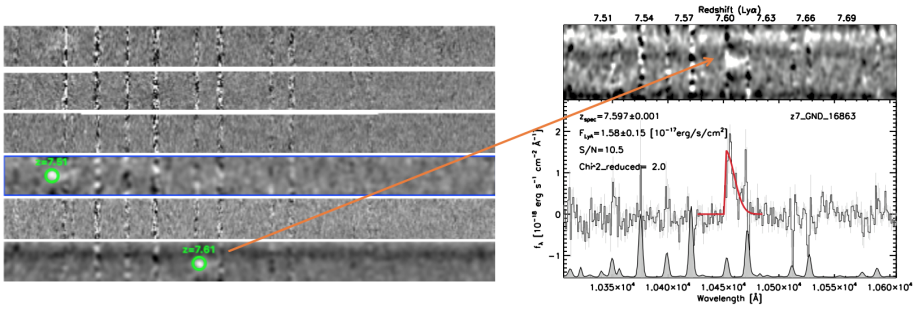

FIGURE 4. (Left) Our MOSFIRE dataset provides the deepest (> 16hrs) NIR spectroscopic data for six candidate galaxies at $z > 7$. The plot shows the fully-reduced and all-mask-combined 2D spectra of the six objects, and we detect two Lyα emission lines at $z = 7.51$ (reported in Finkelstein et al. 2013) and $z = 7.6$ (a new discovery) among the six targets. (Right) 2D and 1D spectrum of the new Lyα detection at $z = 7.6$. The red line shows the asymmetric Gaussian profile of the Lyα line.